CS 241 Data Organization using C
Lab 3: getbits

Joseph Haugh

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1 Description

Section 2.9 of Kernighan and Ritchie provides the `getbits` function. Write a C program that reads records from the standard input stream, passes the data in each valid record to `getbits` and displays the result.

For this program, a record is a sequence of colon delimited numeric characters terminated with a newline. A valid record has the form: `x:p:n`
where x, p, and n correspond to the `x`, `p`, and `n` in `getbits` and where each of these values is within the range usable by `getbits` on a machine (such as `moons.cs.unm.edu`), where an `unsigned int` is 4 bytes.

For each valid input record, you will print a line with the arguments to and the result from calling `getbits`. For each error record, output an error message. There will be sample input and output files for testing on the course web site.

When trying to determine a valid range for n and p, please note that if the right hand operand of shift operator is greater than or equal to the number of bits of the type of the left hand operand, the behavior is undefined. There is one particular error case in the test file that trips up students every year because of this.

2 Error Handling

You need to handle the following errors:

- The value for x is out of range for an unsigned int.
- The value for p is too many bits for an unsigned int.
- The value for n is too many bits for an unsigned int.
- Too many bits are requested from the position.
- A character other than a digit (0-9), a colon, a newline or EOF is encountered.

3 Turning in your assignment

Attach your program file **getbits.c** into the Lab 3 assignment in Canvas.

4 Grading Rubric (total of 20 points)

[-2 points]:	The program does not start with a comment stating the students first and last name and/or the source file is not named correctly.
[-2 points]:	Program compiles with warnings on `moons.cs.unm.edu` machine using `/usr/bin/gcc` with the **-Wall -ansi -pedantic** options
[4 points]:	Code follows the CS-241 Coding Standard
[16 points]:	Passes 16 unknown test cases. For each error case, print the appropriate error message. For each valid test case, print a record of the form: `getbits(x=127, p=3, n=4) = 15`

As before, your program's output will be compared to the expected output using the `diff` utility. Make sure your output matches exactly. Punctuation, capitalization, and whitespace all matter here!

For examples:

```
$ ./a.out < testgetbits.in > myOut.txt
$ diff myOut.txt testgetbits.out
```

Remember you are reading from standard in (getchar) not from a file!